

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1731

Roanoke Rapids 138 kV
January 26, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1731. The Interconnection Customer proposes a 104.0 MW battery energy storage facility interconnecting at the Roanoke Rapids 138 kV substation in the North Carolina area. The request was received on May 01, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 462 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$915,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Cedar Falls Wi - N Tower Tap 138kV L	N Tower Tap - Haggin 345kV	126.5	304	6.4
Mickens - Igo 345kV	Newport News - Mickens 345kV L	102.9	1,089	8.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N Tower Tap - Cedar Falls Wi 138kV	Rebuild line with higher-capacity conductor	\$15,130,000	4% (\$589,000)
Mckens - Igo 345kV Line	Reconductor line	\$18,194,000	2% (\$326,000)
	Total allocated to project		\$915,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.